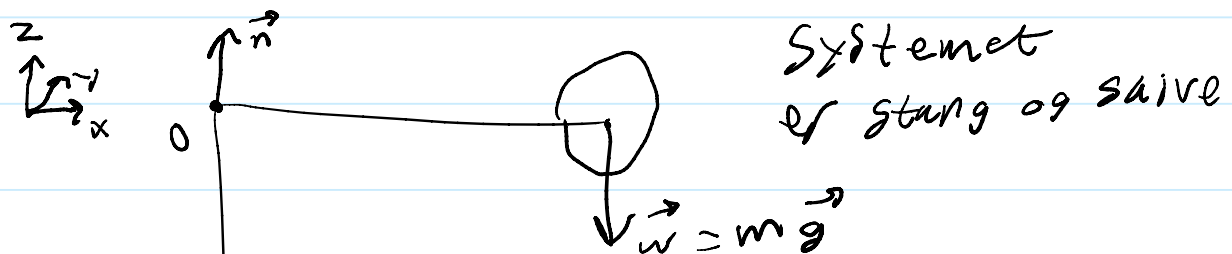


Gyroskop og præcision

Tuesday, 10 October 2023 11.09

Man står med strakt arm
med et hjul i hånden,
Hvis der er en der holder
hjulet oppe og så giver
slip, så ser vi på hvad
der sker

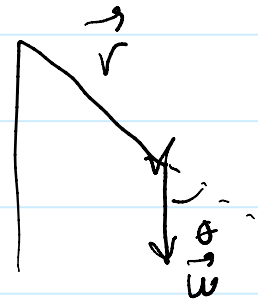


$$\vec{\tau}_0 = \vec{0} \times \vec{n} + \vec{r} \times \vec{w} \quad (\text{i y-aksen})$$

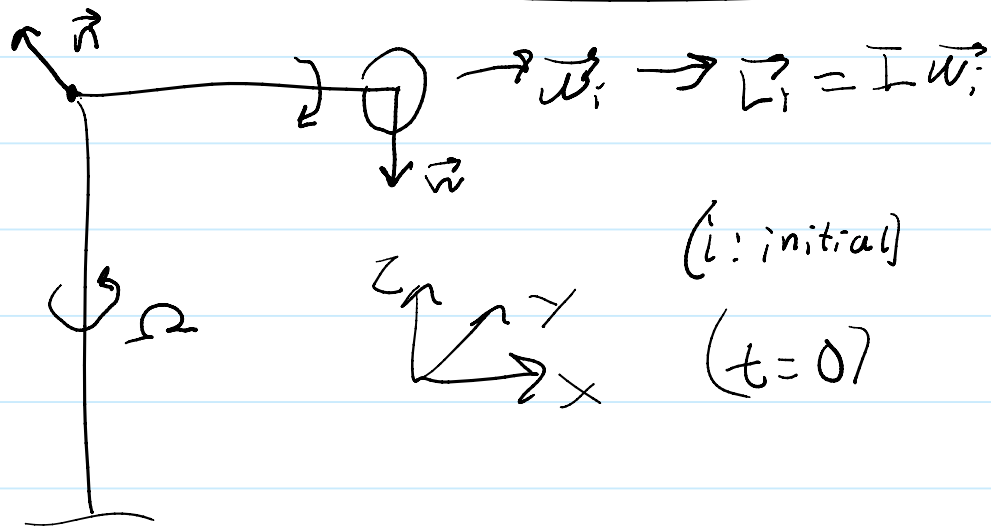
$$\vec{L}_i = \vec{0}$$

Startsituation ingen rotation
af hjul

$$d\vec{L}_i = \vec{\tau}_i dt$$

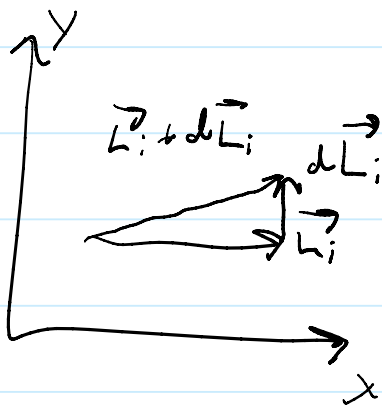


$$\tau = r w \sin \theta$$



$$\vec{\tau}_i = \vec{r}_i \times \vec{w}_i \quad (\text{y-aksen})$$

$$d\vec{L}_i = \vec{\tau}_i dt \quad \vec{L}_i \text{ (x-aksen)}$$



$d\vec{L}_i$ vinkelret på $\vec{L}_i$

Hvilket betyder at rotationsaksen flytter sig i x, y plan

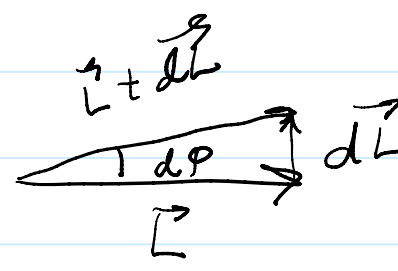
$$\vec{L} \perp \vec{\tau} \Rightarrow \vec{L} \perp d\vec{L}$$

$$\begin{aligned}
 \frac{dL^2}{dt} &= \frac{d}{dt} (\vec{L} \cdot \vec{L}) \\
 &= 2 \cdot \vec{L} \cdot \frac{d\vec{L}}{dt} \\
 &= 2 \vec{L} \cdot \vec{\tau} = 0
 \end{aligned}$$

L ændre sig ikke i tid

Precessionsvinkel hastighed

$$\Omega = \frac{d\phi}{dt}$$

$$\tan \phi = \frac{|\vec{dL}|}{|\vec{L}|} = \frac{\sin d\phi}{\cos d\phi} \sim \frac{d\phi}{1}$$


$$\Omega = \frac{|\vec{dL}|}{dt \cdot |\vec{L}|} = \frac{\tau}{I \omega} = \frac{r m g}{I \omega}$$

Så hvis ω er lille er Ω stor